JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 91, NO. B5, PAGES 4955–4970, APRIL 10, 1986

Paleomagnetism of the Late Proterozoic Tsezotene Formation of Northwestern Canada

JOHN K. PARK

Division of Seismology and Geomagnetism, Earth Physics Branch, Ottawa, Ontario, Canada

JAMES D. AITKEN

Institute of Sedimentary and Petroleum Geology, Geological Survey of Canada, Calgary, Alberta

Paleomagnetic evidence from 37 sites of the partly red-pigmented siliciclastic Tsezotene Formation supports a recently proposed apparent polar wander path for the Hadrynian Mackenzie Mountains supergroup (MMs). The probable primary remanence has a direction at $D°$, $I° = 271$, $+24$ ($k = 15$; $\alpha_{95}° = 8$) with an associated pole T_A (12°N, 214°W; $N = 23$ specimens; $\delta p°$, $\delta m° = 5, 9$). T_A becomes the oldest pole from the MMS. It helps bridge the gap between the apparently youngest poles of the Grenville Loop (about 0.88 Ga) and the suggested younger poles from the MMs. A secondary pole T_B (23°N, 198°W; $N = 18$ sites; $\delta p°$, $\delta m° = 3, 5$), derived from a magnetic direction in hematite pigment ($D°$, $I° = 263$, $+48$; $k = 73$; $\alpha_{95}° = 4$), supports a magnetization found in the overlying "Copper cycle" and in younger units of the MMs as a pervasive overprint. Another secondary pole T_C (63°N, 141°W; $N = 29$ sites; $\delta p°$, $\delta m° = 6, 6$), derived from a magnetization ($D°$, $I° = 317$, $+87$; $k = 89$; $\alpha_{95}° = 3$) partly residing in another phase of hematite pigment, is of postfolding age (post-Paleocene). This study demonstrates the importance of using several treatment methods, singly and in combination, when analyzing complex magnetizations.

INTRODUCTION

Paleomagnetism is a potential dating tool if reference apparent polar wander (APW) paths, calibrated by geological and radiometric methods, have been established [*Park*, 1983]. Sometimes, even without a reference base, paleomagnetic results may be used to constrain ages. Studies of the Precambrian sequence in the Mackenzie Mountains of the Northwest Territories were begun with this in mind.

Until recently, no reliable radiometric dates were available from the region. On the basis of style of sedimentation and distribution, broad correlations were assumed between the Mackenzie Mountains supergroup of the northern Cordillera and the Purcell Supergroup of the southern Cordillera and between the northern Rapitan Group and younger succession of the north (Ekwi supergroup) and the Windermere Supergroup of the south [*King*, 1969; *Douglas*, 1970; *Stewart*, 1972; *Gabrielse*, 1972; *Donaldson*, 1973; *Eisbacher*, 1978, 1981; *Young et al*, 1979]. New information invalidates the assumed correlation of the older supergroups [*Young*, 1979] but supports the correlation of the younger successions [*Evenchick et al.*, 1984].

Young [1979] has demonstrated that the Mackenzie Mountains supergroup is younger than the 1.2 Ga Coppermine lavas. A recalculated date for the Hellroaring Creek stock, which cuts folded Purcell rocks, indicates that deposition, folding, and metamorphism of the classical Purcell were complete by no later than 1.3 Ga [*McMechan and Price*, 1982]. On the other hand, granitic rocks firmly dated on about 0.735 Ga nonconformably underlie the Windermere succession at two localities [*Evenchick et al.*, 1984]. The Rapitan Group of the Mackenzie Mountains postdates basic intrusions dated at about 0.77 Ga [*Armstrong et al.*, 1982], thus the Windermere–Ekwi correlation is supported by the radiometric dates and united by the rifting event to which both supergroups are a depositional response [*Young et al.*, 1979; *Eisbacher*, 1981].

Copyright 1986 by the American Geophysical Union.

Paper number 5B5643.
0148-0227/86/005B-5643$05.00

To constrain further the age of the Mackenzie Mountains supergroup, using paleomagnetism, paleopoles determined from the supergroup were compared to other possibly correlative paleopoles from North America. Precambrian poles with which to compare results are few. An early attempt at correlation suggested a possible relation to poles of the Grenville Loop [*Park*, 1981*a*], dated at about 1000–900 Ma. A later interpretation, based on more results, suggested a younger age range [*Morris and Park*, 1981]. That interpretation proposed accommodating poles of the Mackenzie Mountains supergroup within a younger antipolar loop, connected to the Grenville Loop via high-latitude poles. On the basis of newly determined paleopoles, we have recently suggested an alternative APW path for the Mackenzie Mountains poles [*Park and Aitken*, 1985].

Most poles from the Proterozoic rocks of the region fall west of the Grenville Loop (see Figure 11). The sequence and position of probable primary poles from the Mackenzie Mountains indicated a new polar path, younger than the Grenville Loop (less than 880 Ma [*Dallmeyer and Sutter*, 1980]). The new path is defined by primary poles from the Katherine Group [*Park and Aitken*, 1986] and formations within the overlying Little Dal Group [*Park*, 1981*a*, 1984*a*, and unpublished data, 1984] and by a secondary hematite pigment pole from the lower Little Dal Group [*Park*, 1981*a*].

The Mackenzie Mountains supergroup APW path may terminate at pole S_A from sills dated at about 0.77 Ga that cut the Tsezotene Formation (see Figure 11) [*Armstrong et al.*, 1982; *Park*, 1981*b*]. This interpretation does not depend on the assumption that the sills are contemporaneous with lavas at the top of the Little Dal Group; the marked difference between palepoles for sills and lavas denies that they are coeval [*Morris and Park*, 1981]. The pole S_A lies near a projection of the path from the Little Dal poles LB_A and LR_A (J. K. Park, unpublished data, 1984) to the "Copper cycle" poles CR_A and CB_A (see Figure 11). We interpret these data as evidence that the sills were emplaced during an episode of rifting to which "Copper cycle" and lower Rapitan deposition were the response.

This paper adds a probable primary pole from the Tsezo-

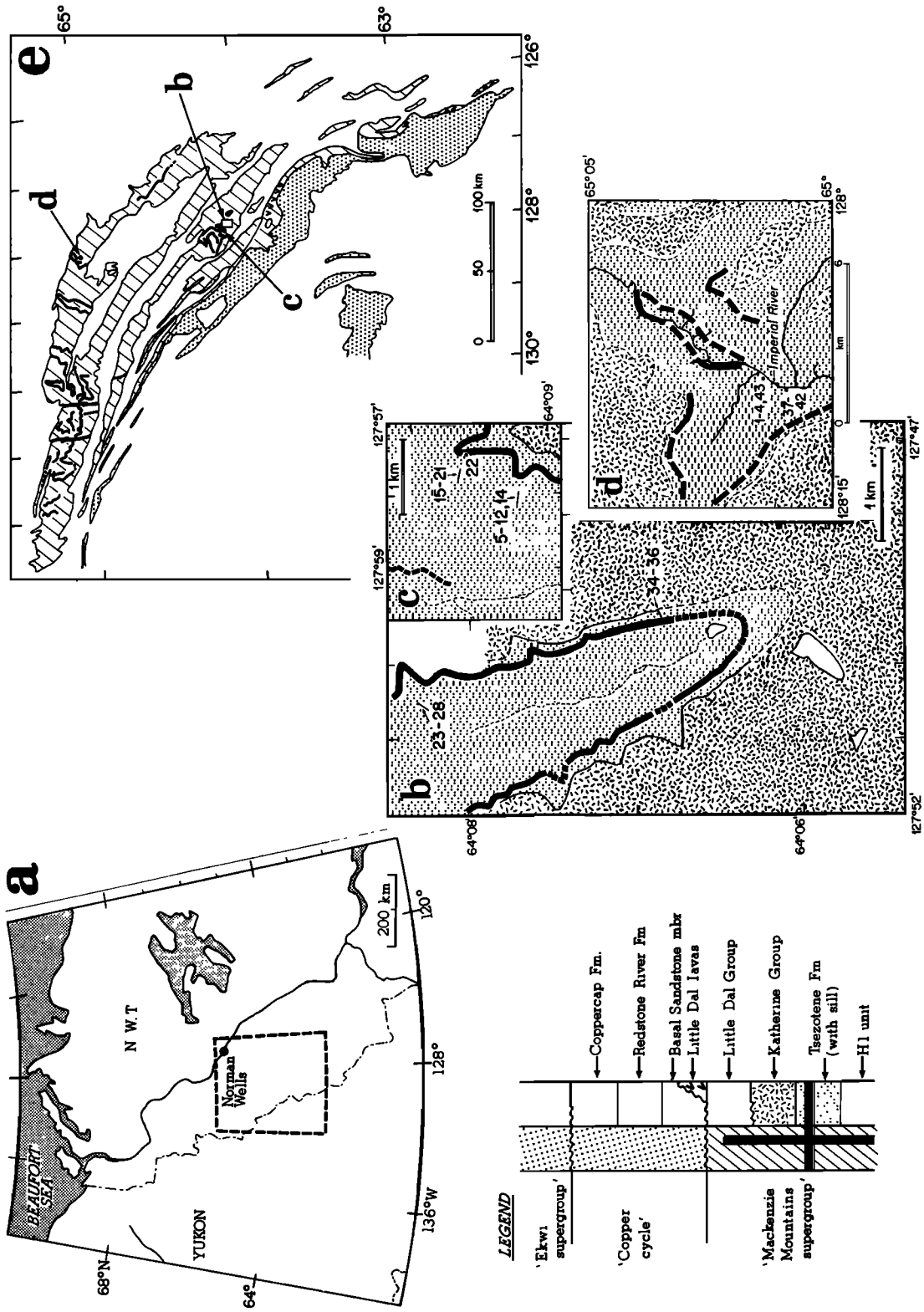


Fig. 1. (*a*) Location and (*b*)-(*e*) generalized geological maps of the Mackenzie Mountains region. Figure 1*e*, the subregion in Figure 1*a*, is after *Armstrong et al.* [1982].

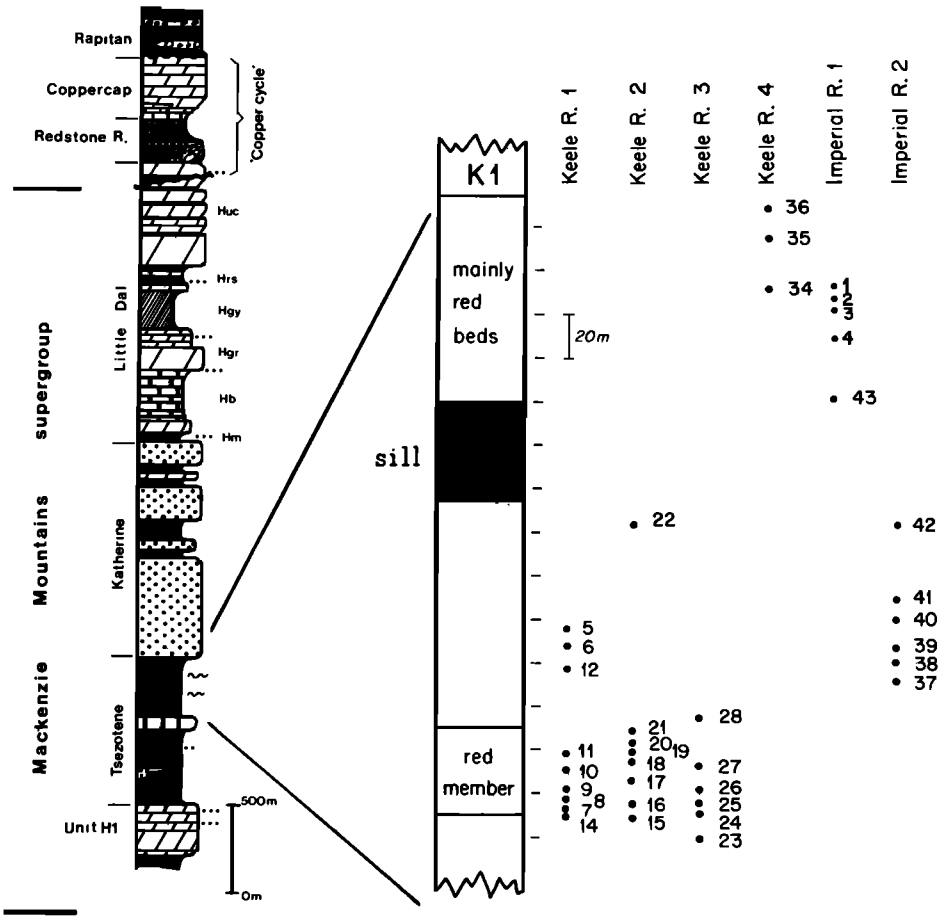


Fig. 2. Stratigraphy of the Mackenzie Mountains supergroup. The approximate stratigraphic heights of sites are shown in the generalized expanded section of the upper Tsezotene Formation. The Little Dal Group is composed of the following units: Upper Carbonate formation (Huc); Rusty Shale formation (Hrs); Gypsum formation (Hgy); Grainstone formation (Hgr); Basinal/Platform assemblages (Hb), and Mudcracked formation (Hm) [*Aitken*, 1981]. Coordinates of the six localities are given in Tables 2–4.

tene Formation to the proposed polar path. It becomes the oldest pole presently available from the Mackenzie Mountains supergroup. It appears to fit into the polar sequence according to age and supports the previous interpretation of a connection between the youngest part of the Grenville Track and the oldest part of the path from the Mackenzie Mountains supergroup [*Park and Aitken*, 1986].

GEOLOGY

The exposed Precambrian rocks in the Mackenzie Mountains consist of two major, regionally continuous successions, the Mackenzie Mountains supergroup and the younger, Ekwi supergroup [*Young et al.*, 1979], separated by a minor and markedly discontinuous succession, the "Copper cycle" [*Young et al.*, 1970; *Aitken*, 1981] (Coates Lake Group of *Jefferson and Ruelle* [1986]) (Figure 1). Because the base of the "Copper cycle" is strongly uncomfortable [*Aitken*, 1981] and the top may be locally conformable [*Jefferson*, 1978], stratigraphers currently involved with the Mackenzie Mountains Proterozoic are treating the "Copper cycle" as more closely related to the Ekwi supergroup than to the underlying Mackenzie Mountains supergroup within which it was originally included.

The Tsezotene Formation of the Mackenzie Mountains supergroup [*Gabrielse et al.*, 1973] appears to be conformable with bounding units [*Aitken et al.*, 1978; *Aitken and Cook*, 1974] (Figure 2). The mainly clastic rocks define a coarsening upward sequence, exceeding a thickness of 1220 m in places. The two regionally recognizable parts of the formation are locally separated by stromatolite bioherms at the base of the upper member. "The lower member is dominated by gray argillaceous rocks and the upper member by varicoloured argillite, sandstone, and minor carbonate" [*Aitken et al.*, 1978, p. 481]. The top of the Tsezotene is intruded, wherever exposed, by one or more diabase sills [*Aitken et al.*, 1982], assumed to be related to dykes of similar composition that intrude the Mackenzie Mountains supergroup as high as the middle of the Little Dal Group.

Paleomagnetic samples were collected from the upper member in two regions of the Mackenzie Mountains (Figures 1 and 2). Most of the sites are from red-pigmented fine- to medium-grained sandstone.

METHODS

Rocks were oriented by sun and magnetic compasses prior to hand sampling. In the laboratory, samples were cored, and cores sliced into from two to four specimens of mostly standard size.

Following measurement of the natural remanent magnetization (NRM), specimens were subjected to one of three demagnetization treatments: alternating field (af), thermal, or chemical. Sometimes a second treatment was applied, that is,

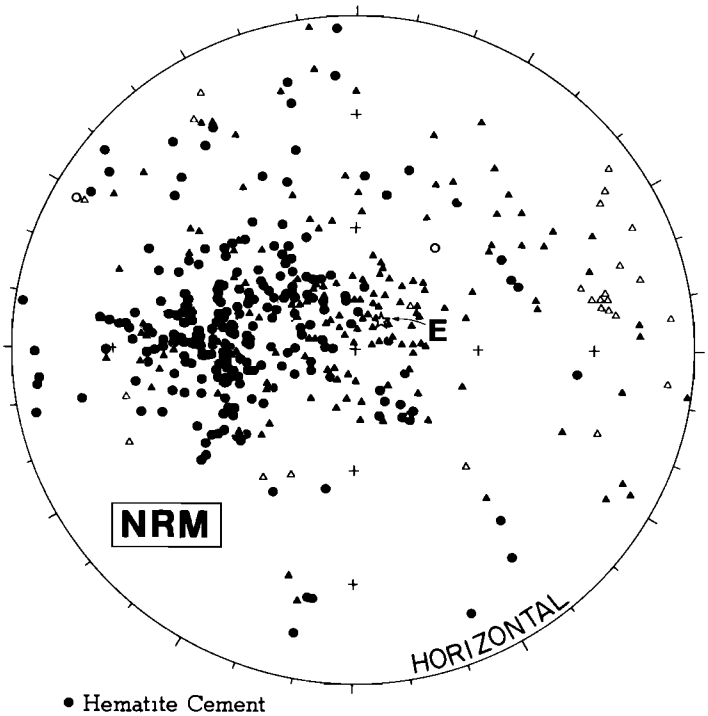


Fig. 3. Natural remanent magnetization (NRM) directions of individual specimens. Solid symbols are downward directed; open symbols, upward directed. E represents the direction of the present earth's field in the sampling region.

thermal might follow af or chemical treatment, or af might follow chemical treatment. Treatments were continued until remanent magnetizations became too weak or were masked by induced magnetizations.

The af treatments were done with a three-axis tumbler [*Roy et al.*, 1973] in residual fields of less that 2 nT. Peak fields were increased from 5 to a maximum of 100 to 290 mT, in increments of 5–50 mT. Decay times from peak field ranged from 30 to 300 s. Thermal demagnetization up to 720°C was done with commercial (Schonstedt TSMs) and in-house instruments [*Roy et al.*, 1972]. Thermal treatments were usually in increments of 50°–100°C for the first 500°C and from 10° to 20° above 500°C, depending on the maximum unblocking temperatures (T_{UB}). Specimens were generally alternately inverted in the oven, so that the acquisition of any induced partial thermal remanent magnetization (PTRM) could be gauged. Residual fields in the noncommercial oven were adjusted to within 2 nT prior to heating runs [*Roy et al.*, 1969].

Chemical treatment was done according to the procedures of *Park* [1970]. Specimens were drilled with an axial hole to facilitate leaching. They were then submerged in hydrochloric acid for periods ranging from 7 hours to 28 days, withdrawn from acid, washed in water, and measured. Acid strengths ranged from 6 to 12 *N*. Lower acid strengths and shorter leaching times were used initially to inhibit disintegration and to provide more measurement data for specimens whose remanence intensities decayed rapidly. Accumulated leaching times ranged to a maximum of about 7000 hours.

Three separate chemical demagnetization experiments were conducted over 5 years, during which the general technique of leaching was supplemented. Several techniques were introduced to speed up the turnaround time for measurement. Usually before measurements were made, specimens were stored in magnetic shields, so that any viscous remanent magnetization (VRM) built up during leaching would decay. During the second experiment, specimens were treated in an af of 5 mT after each leaching step to produce the same result. This procedure was discontinued because components of both VRM and NRM were removed and the effect on the NRM could not be adequately gauged. In the third experiment, specimens were leached under vacuum in a reduced field environment (about 12% of the earth's field). The vacuum reduced leaching times by about half. Another useful technique consisted of inverting specimens from one leaching step to the next. This served to randomize and reduce any possible VRM or chemical remanant magnetization (CRM) produced during the experiment [*Roy and Park*, 1974; *Henshaw and Merrill*, 1980].

The three demagnetization techniques, singly and in combination, were necessary to analyze the complex NRM. Often components were found to have overlapping T_{UB}, resistive coercive force (RCF), and leaching characteristics. Components were calculated from lines drawn by eye through apparent straight-line segments of vector diagrams. Determinations from the noisy vector representations of weakly magnetized specimens were more subjective. Specimen rather than sample directions are used to calculate site mean directions and statistics [*Park and Aitken*, 1986].

Results and Analysis

NRM directions for specimens containing visible hematite cement tend to cluster toward the west with moderate to steep

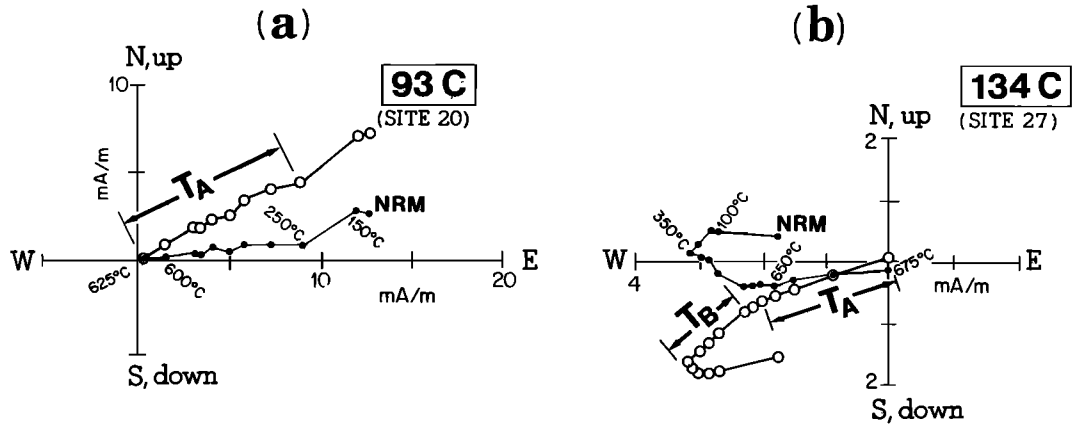


Fig. 4. Vector representations of remanent magnetization during stepwise thermal demagnetization of specimens from sites (*a*) 20 and (*b*) 27. Solid circles represent data projected onto the horizontal (N-E) plane; open circles, data projected onto the east-west vertical (up-E) plane. The approximate removal ranges of T_A and T_B are indicated.

TABLE 1. Characteristics of Magnetic Components

Treatment Type	T_A (sp)	T_B (sp)	T_C (sp)
	Keele River 1 (5, 6, 12, 11, 10, 9, 8, 7, 14)		
th	682°–700°C (1)	50°–685°C (14)	20°–550°C (6) 20°–650°C (2)
af	?	?	0°–>80 mT (4)
af + th	?	40°–690°C (2)	20°–600°C (3)
ch	553–4284 hours (1)	0–4956 hours (23)	24–2436 hours (16)
ch + th	?	350°–>640°C (1)	100°–600°C (13)
ch + af	?	?	5–>40 mT (13)
	Keele River 2 (21, 20, 19, 18, 17, 16, 15)		
th	550°–674°C (4)	<450°–675°C (3)	50°–550°C (5) 20°–650°C (4)
af	0–40 mT (2)	?	10–100 mT (2)
af + th	?	<690–710°C (1)	20°–695°C (4)
ch	24–6174 hours (2)	0–2425 hours (11)	48–1056 hours (5)
ch + th	20°>575°C (2)	?	20°–550°C (2)
ch + af	0–25 mT (1)	?	0–>40 mT (4)
	Keele River 3 (28, 27, 26, 25, 24, 23)		
th	600°–685°C (4)	300°–677°C (6)	50°–575°C (4) 350°–620°C (1)
af	?	?	?
af + th	?	20°–710°C (4)	100°–690°C (2)
ch	?	0–4284 hours (15)	0–769 hours (9)
ch + th	?	?	20°–>550°C (6)
ch + af	?	?	0–>40 mT (6)
	Keele River 4 (36, 35, 34)		
th	?	550°–674°C (2)	50°–500°C (1)
af	0–40 mT (1)	?	0–90 mT (3)
af + th	?	20°–700°C (2)	?
ch	?	0–2425 hours (4)	?
ch + th	?	?	200°–550°C (2)
ch + af	?	?	5–>30 mT (30)
	Imperial River Canyon 1 (43, 1, 2, 3, 4)		
th	?	...	20°–450°C (5)
af	?	...	10–50 mT (3)
af + th	?	...	20°–500°C (2)
ch	?	...	0–2438 hours (8)
ch + th	250°–500°C (2)	...	20°–500°C (3)
ch + af	1–>40 mT	...	2.5–10 mT (1)
	Imperial River Canyon 2 (42, 41, 40, 39, 38, 37)		
th	?	250–680°C (9)	20°–550°C (16) 350°–660°C (1)
af	?	?	0–>250 mT (3)
af + th	?	250°–680°C (5)	20°–575°C (8) 665°–700°C (1)
ch	?	96–2425 hours (12)	0–696 hours (17)
ch + th	?	?	20°–>550°C (2)
ch + af	?	?	2.5–>40 mT (5)

Specimens (sp) are given in parentheses; af, alternating field; ch, chemical leaching; th, thermal; +, followed by.

positive (downward) inclinations (Figure 3). Intensities range from 0.1 to 90 mA/m. The considerable intensity variation within sites suggests mixed magnetic phases and/or uneven distribution of phases.

Thermal demagnetization revealed, on the basis of direction, three magnetic components, but not always in the same specimen. Often the components have largely overlapping T_{UB} and cannot be clearly resolved. Two of the components, called T_A and T_B, are illustrated in Figure 4. Each has a westerly, positively inclined direction, but the inclination of T_A is slightly shallower. T_A may also be reversed or easterly directed (Figure 4*a*). The two components have similar T_{UB} that are consistent with a hematite carrier; though where separation can be achieved, T_A tends to have slightly higher T_{UB} (Table 1). A third component T_C, with steep positive inclination, has T_{UB} consistent with both magnetite and hematite.

The af treatment mainly resolved only the steep T_C component, but in three instances it revealed T_A (Table 1). Most RCFs are below 100 mT, consistent with a magnetite carrier, but several specimens have RCFs greater than 250 mT, indicating a probable hematite carrier.

Chemical leaching revealed all three components uncovered by thermal and af treatments. T_C is often resolved first (Figures 5 and 6*b*) in leaching periods up to 2438 hours (Table 1). T_B is usually resolved next and is the most commonly determined component. At locality 1 in the Imperial River Canyon, T_B is missing, and T_C is removed within the same leaching interval as T_B at the Keele localilites (Table 1). T_C is evidently associated with the red pigment observed in Imperial samples. Therefore, at least two distinct phases of pigmentation coexist in the Tsezotene Formation, with T_C, removed first at Keele River, being the latest phase. T_A is resolved by leaching in

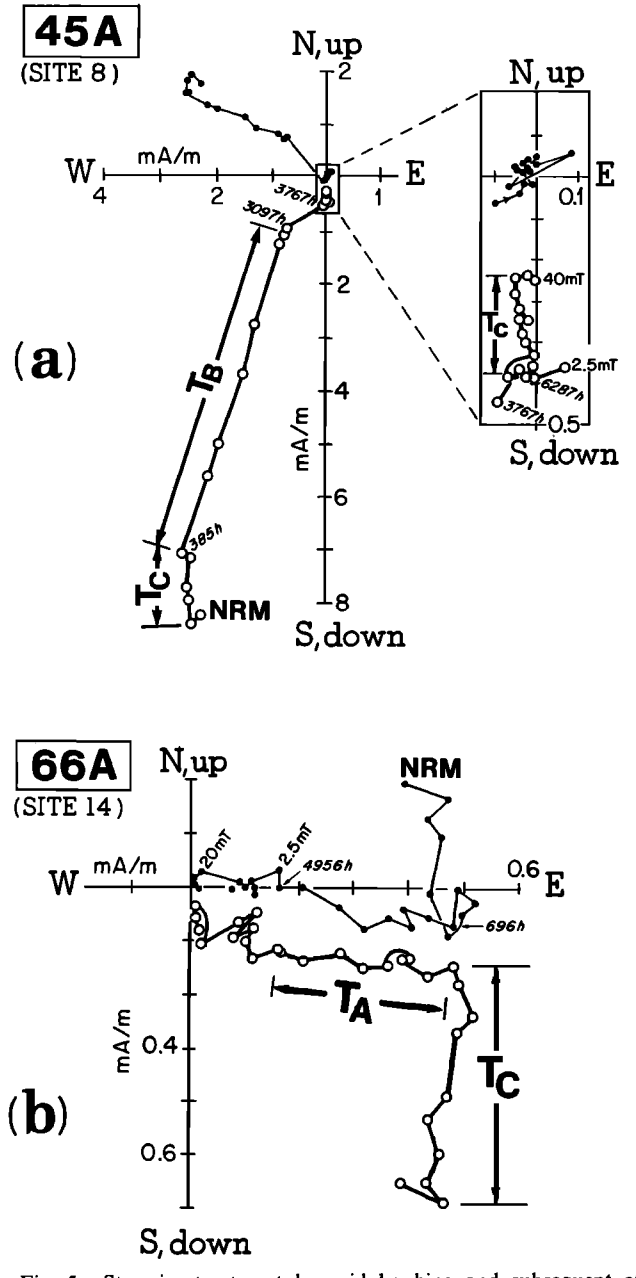


Fig. 5. Stepwise treatment by acid leaching and subsequent af treatment of specimens from sites (*a*) 8 and (*b*) 14. Approximate removal ranges for magnetic components T_A, T_B, and T_C are shown. See Figure 4 for other explanations.

only three instances. In each case, the specimen had little or no initial pigment (Figure 5*b*). At site 8, where abundant red pigment is present, microscopic examination of representative leached and unleached specimens revealed a lesser amount of detrital hematite in the leached specimens, indicating that leaching probably removed some of the detrital hematite in addition to the pigment. This observation correlates with remanence mesurements from at least one site 8 specimen (Figure 5*a*), which reveal the probable removal of T_A between 3097 and 3767 hours following the elimination of T_B and T_C with the pigment.

Most of the chemically treated specimens were later af or thermally demagnetized to determine whether the remanent magnetization was single or composite. It was often found to be composite, consisting of another T_C phase (Figures 5*a* and 6) and a soft magnetization with RCFs under about 5 mT and T_{UB} less than about 250°C (Figure 6*a*). Thermal and af treatments of chemically leached specimens always reveal the second T_C phase to have T_{UB} or RCFs consistent with a magnetite carrier (Figure 6), but thermal treatment of unleached specimens, in addition, also indicates a hematite carrier (Table 1). Evidently, chemical leaching had eliminated the T_C hematite phase revealed under thermal demagnetization alone; therefore the T_C pigment is hematite.

Some af specimens were later treated thermally; again as in the chemical case, to determine whether the remanent magnetization was single or composite. Thermal treatment revealed either T_B or T_C and in four cases, both T_B and T_C. As under thermal demagnetization above, the T_{UB} of T_C are indicative of both magnetite and hematite carriers (Table 1). The T_{UB} of T_B tend to be higher.

The T_A, T_B, and T_C directions are set out according to site in Tables 2, 3, and 4, respectively. Individual directions of T_C are plotted according to treatment in Figure 7 and are summarized in Table 5. Because of the similarity of nominated T_C directions under different treatments (rows 1–6) and the fact that all directions tend to be postfolding (see below), they have been combined within sites (Table 4 and Figure 8). (Grand averages are given in Table 5 (rows 7–15)). Because of directional agreement under different treatments (Table 6, rows 1-3), nominated T_B directions have also been combined within sites (Table 3 and Table 6, rows 4–11). Individual T_B directions are plotted in Figure 9 according to treatment, and site directions are plotted in Figure 8. T_A directions are very scarce and are almost equally divided between normal and reverse polarity (Table 7 and Figure 10). These directions are most common in the reverse site 20, which contains only a trace of hematite pigment. In this site, the good agreement among directions determined by almost every technique indicates that all T_A directions should be averaged together. The T_A direction is supported by vector subtraction directions calculated from high-temperature (>620°C) data (Table 7). Directions were obtained by (1) subtracting presumed end point values, which are either normal or reverse, from the respective reverse or normal vectors present prior to reversal, or (2) averaging a consistent vector substraction direction removed over a range of temperatures (technique of *Park and Roy* [1979] and *Park* [1981*a*]).

A tilt test was applied to all the directions (Table 8). The precision of the averages was marginally improved for T_A and T_B after correction for bedding; however, the test is not significant to prove that these directions are of prefolding age. But for T_C the test is significant. In fact, for the Imperial River sites, where the two localities are on opposite limbs of a fold, a negative fold test is significant at the 99% probability level to prove that T_C is of postfolding age (post-Paleocene) [*Aitken et al.*, 1982]. Both T_A and T_B have directions that are far different from Paleocene or later directions expected for the region; therefore they are most probably older and thus prefolding. In fact, T_B is similar to the prefolding K_B magnetization of the overlying Katherine Group [*Park and Aitken*, 1986].

DISCUSSION AND CONCLUSIONS

Nature of the NRM Components

Microscopic examination of polished thin sections revealed that most specimens contain detrital and pigmental hematite. Specimens from some sites (2–4, 10) contain no discernible detrital grains; specimens at other sites (12, 14, 15, 43) contain no pigment. Many sites contain only a trace of pigment (1–3,

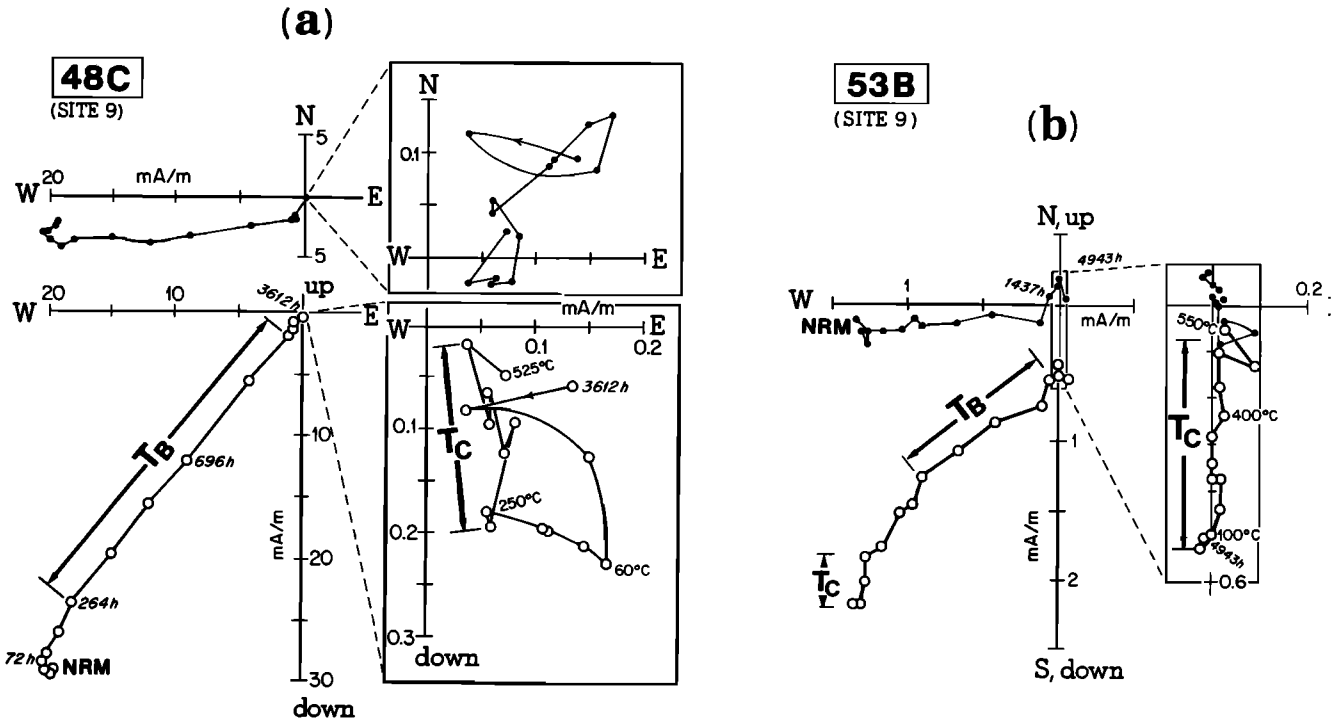


Fig. 6. Stepwise treatment by acid leaching and subsequent thermal treatment of specimens from site 9. See Figure 4 for other explanations.

TABLE 2. T_A Directions and Poles

			Directions						Poles (Bedding)	
			Horizontal		Bedding					
Site	N	Treatment	$D°$	$I°$	$D°$	I°	R	$\alpha_{98}{}^0$	°N	°W
Keele River Section 1 (64.2°N, 128.0°W)										
5, 6, 12-8	...	...	...	...	...	...	...	...	...	...
7	1	th	287	+16	285	+18	...	...	15	228
14	1	ch	102	+07	102	+05	...	...	03	230
Keele River Section 2 (64.2°N, 128.0°W)										
21	1	th	280	−02	280	+03	...	...	06	226
20	8	th, af, ch, ch + th	080	−27	083	−28	7.90	7	10	205
19-17	...	...	...	...	...	...	...	...	...	...
16	1	ch + af	312	+38	311	+38	...	...	36	247
15	...	...	...	...	...	...	...	...	...	...
Keele River Section 3 (64.1°N, 127.8°W)										
28	...	...	...	...	...	...	...	...	...	...
27	1	th	264	+24	265	+24	...	...	09	208
26	2	th	276	+22	276	+28	1.99	...	16	217
25	1	th	260	+27	261	+35	...	...	13	201
24, 23	...	...	...	...	...	...	...	...	...	...
Keele River Section 4 (64.1°N, 127.8°W)										
36, 35	...	...	...	...	...	...	...	...	...	...
34	1	af	080	−10	079	−09	...	...	−01	206
Imperial River Section 1 (65.0°N, 128.1°W)										
43	...	...	...	...	...	...	...	...	...	...
1	1	ch + af	287	+25	290	+31	...	...	24	229
2	1	ch + th	304	−31	299	−28	...	...	−02	250
3	2	ch + th/af	268	+24	273	+31	1.94	...	11	211
4	...	...	...	...	...	...	...	...	...	...
Imperial River Section 2 (65.0°N, 128.2°W)										
42	1	af	262	+56	239	+29	...	...	01	184
41, 42	...	...	...	...	...	...	...	...	...	...
39	1	af	115	−44	091	−37	...	...	19	210
38, 37	...	...	...	...	...	...	...	...	...	...

N is the number of specimens. $D°$, $I°$ is the declination and inclination of the direction with respect to either the present horizontal and true north or the bedding; R, the vector resultant; and $\alpha_{95}°$, the half angle of the cone of 95% confidence about the direction. When two treatments on the same specimen reveal the same direction, the directions are averaged to produce the specimen direction. Other symbols are as explained in Table 1.

TABLE 3. T_B Directions and Poles

Site	N	Treatment	Directions: Horizontal D°	Horizontal I°	Bedding D°	Bedding I°	R	α_{98}°	Poles (Bedding) °N	Poles (Bedding) °W
			Keele River Section 1 (64.2°N, 128.0°W)							
5	6	ch, th, ch + th	245	+44	243	+41	5.91	9	10	184
6, 12	...	...	...	...	...	...	...	...	...	...
11	4	ch, th	262	+46	258	+47	3.81	24	21	195
10	...	...	...	...	...	...	...	...	...	...
9	8	ch, th, af + th	264	+48	258	+45	7.95	5	19	196
8	2	ch, th	264	+39	261	+41	1.97	...	17	199
7	8	ch, th, af + th	267	+50	260	+51	7.65	13	24	194
14	2	ch, th	267	+40	266	+40	1.97	...	19	204
			Keele River Section 2 (64.2°N, 128.0°W)							
21, 20	...	...	...	...	...	...	...	...	...	...
19	5	ch, th	270	+53	267	+50	4.97	6	26	201
18	6	ch, af + th	258	+52	256	+52	5.86	11	23	191
17	...	...	...	...	...	...	...	...	...	...
16	4	ch, th	272	+46	271	+46	3.94	13	25	206
15	...	. .	...	...	...	...	...	...	...	...
			Keele River Section 3 (64.1°N, 127.8°W)							
28	1	af + th	295	+52	300	+49	...	...	(40)	(231)
27	3	ch, th	281	+44	285	+47	2.93	23	32	218
26	7	ch, th, af + th	267	+43	266	+49	6.89	8	25	200
25	3	ch, th	270	+54	276	+61	2.98	13	40	200
24	11	ch, th, af + th	276	+56	278	+60	10.91	4	39	203
23	...	...	...	...	...	...	...	...	...	...
			Keele River Section 4 (64.1°N, 127.8°W)							
36	2	ch, af + th	267	+37	266	+37	2.00	4	17	205
35	6	ch, th, af + th	268	+44	264	+47	5.78	14	22	199
34	...	...	...	...	...	...	...	...	...	...
			Imperial River Section 1 (65.0°N, 128.1°W)							
43, 1-4	...	...	...	...	...	...	...	...	...	...
			Imperial River Section 2 (65.0°N, 128.2°W)							
42	...	...	...	...	...	...	...	...	...	...
41	9	ch, th, af + th	290	+64	250	+50	8.84	7	20	188
40	3	ch, af + th	293	+65	248	+51	2.96	17	20	186
39	14	ch, th, af + th	289	+51	262	+40	13.84	4	17	202
38, 37	..	. .	..	...	...	...	...	...	...	...

Symbols as defined in Tables 1 and 2. Those poles defined from only one specimen (values in parentheses in last column) have been excluded from further averaging (Table 6).

16, 20, 23, 34–38). In some specimens the red pigment comprises two generations; the early one coats individual grains, and the later one fills fractures that extend through ground mass and lithic fragments. Magnetite is identified only sporadically, for example, at site 20, where one grain was observed to be partly altered to hematite. It may also occur in felsic and mafic silicates along with hematite, though small included opaques could not be identified, except for occasional hematite grains. Other opaques observed include minor sulphides.

During acid leaching, T_A is removed following the elimination of T_B and/or T_C carried by the red pigment; this suggests that T_A must occur in the detrital hematite grains. Further support for this contention comes from the common occurrence of T_A at the reverse site 20, which has only a trace of pigment, and from the presence of T_A at the normal site 14, which contains no pigment. No other component is revealed after T_A by acid leaching up to 7000 hours, despite a dramatic reduction in the size and abundance of detrital hematite grains; therefore T_A is the major, if not the only magnetization carried by detrital grains. T_A may additionally be carried by exsolved/included hematite in the silicates. In the absence of contrary evidence we suggest that T_A is primary [*Park and Aitken*, 1986].

T_B and T_C are both carried in red pigment, and T_C additionally in a phase revealed by af or thermal treatment after acid leaching. T_B definitely resides in a hematite phase as the earlier of the two components. A fold test applied to the similar K_B magnetization of the overlying Katherine Group indicates that T_B is of prefolding age. The T_C pigment and postleaching phases both appear to postdate the folding. Because of their similar direction and timing they are treated as one component. The red pigment phase of T_C, like T_B, also appears to reside in hematite. Its T_{UB} fall into the ranges indicative of magnetite and hematite, but only the magnetite range remains after acid leaching, thus supporting a hematite carrier for the pigment phase.

The postleaching phase of T_C must dwell in a mineral largely unaffected by acid leaching. It may therefore reside in the submicroscopic magnetite in some of the mafic and felsic silicate grains. This postfolding direction is clearly the result of either chemical or thermal overprinting. Chemical overprinting is suggested by the presence of minor sulphides at some sites. These indicate a reducing environment, which at some time could have converted hematite to magnetite. But the lack of magnetite in the detrital hematite grains accessible to acid argues against a chemical remanence. Instead, we pro-

TABLE 4. T_C Directions and Poles

Site	N	Treatment	Direction: Horizontal D°	Direction: Horizontal I°	Direction: Bedding D°	Direction: Bedding I°	R	α_{98}°	Poles (Horizontal) °N	Poles (Horizontal) °W
			Keele River Section 1 (64.2°N, 128.0°W)							
5	5	ch, th, ch + th/af	220	+78	215	+75	4.83	16	44	149
6	1	af	278	+75	266	+78	...	...	(55)	(183)
12	...	...	...	...	...	...	...	...	...	...
11	7	ch, th, af + th, ch + th/af	299	+80	281	+83	6.88	9	67	175
10	...	...	...	...	...	...	...	...	...	...
9	5	af, ch + th/af	007	+83	318	+88	4.98	5	78	120
8	13	af, ch, th, af + th, ch + th/af	094	+86	120	+83	12.76	6	62	109
7	7	ch, th, ch + th/af	163	+87	169	+81	6.71	14	58	124
14	4	ch, th, af + th, ch + th/af	051	+75	065	+76	3.84	21	67	054
			Keele River Section 2 (64.2°N, 128.0°W)							
21	1	th	169	+83	151	+79	...	...	(51)	(124)
20	2	af, ch + af	270	+89	002	+85	2.00	...	64	130
19	2	ch, th	235	+84	233	+81	1.99	...	57	144
18	8	ch, th, af + th, ch + th/af	308	+89	258	+89	7.94	5	66	133
17	3	af, ch, ch + af	340	+84	292	+88	2.97	15	75	145
16	2	th, ch + th	042	+81	045	+82	1.97	...	73	086
15	3	th	356	+84	008	+81	2.98	11	76	131
			Keele River Section 3 (64.1°N, 127.8°W)							
28	6	ch, af + th, ch + th	319	+74	327	+69	5.84	12	70	212
27	2	ch, th	315	+66	326	+65	1.82	...	(61)	(230)
26	4	th, ch + th/af	274	+81	279	+88	3.98	8	61	163
25	6	ch, th, ch + th/af	275	+80	324	+85	5.69	17	59	169
24	5	th, ch + th/af	313	+87	029	+87	4.98	6	68	140
23	2	af + th, ch + th	047	+73	052	+66	1.89	...	67	044
			Keele River Section 4 (64.1°N, 127.8°W)							
36	1	ch + af	281	+80	269	+80	...	...	(61)	(170)
35	4	th, ch + th/af	294	+77	275	+80	...	...	63	186
34	3	af	042	+82	068	+84	...	...	73	089
			Imperial River Section 1 (65.0°N, 128.1°W)							
43	4	af, th, af + th	241	+74	281	+85	3.61	36	43	165
1	4	ch, th, ch + th	282	+85	030	+85	3.94	13	65	150
2	6	ch, th	224	+90	050	+80	5.95	7	65	129
3	5	ch, th, af + th, ch + th/af	265	+81	346	+81	4.94	10	58	164
4	2	af, ch, ch + th	262	+82	010	+77	1.98	36	59	159
			Imperial River Section 2 (65.0°N, 128.2°W)							
42	2	th	044	+53	116	+80	1.82	...	(48)	(010)
41	10	ch, th, af + th, ch + th/af	359	+89	209	+62	9.80	7	66	128
40	7	af, ch, th, af + th	352	+81	223	+69	6.66	15	82	145
39	11	af, ch, th, af + th, ch + th/af	236	+83	213	+56	10.65	9	56	148
38	11	af, ch, th, af + th, ch + th/af	040	+81	202	+70	10.89	5	74	084
37	5	ch, th, af + th	324	+85	219	+64	4.96	7	73	149

Symbols as defined in Tables 1 and 2. Poles in parentheses in the last two columns have not been used in the determination of the final averages (Table 5). Excluded poles are those defined from only one specimen or from two specimens that have poor directional agreement (arbitrarily, a difference in direction of greater than 40°).

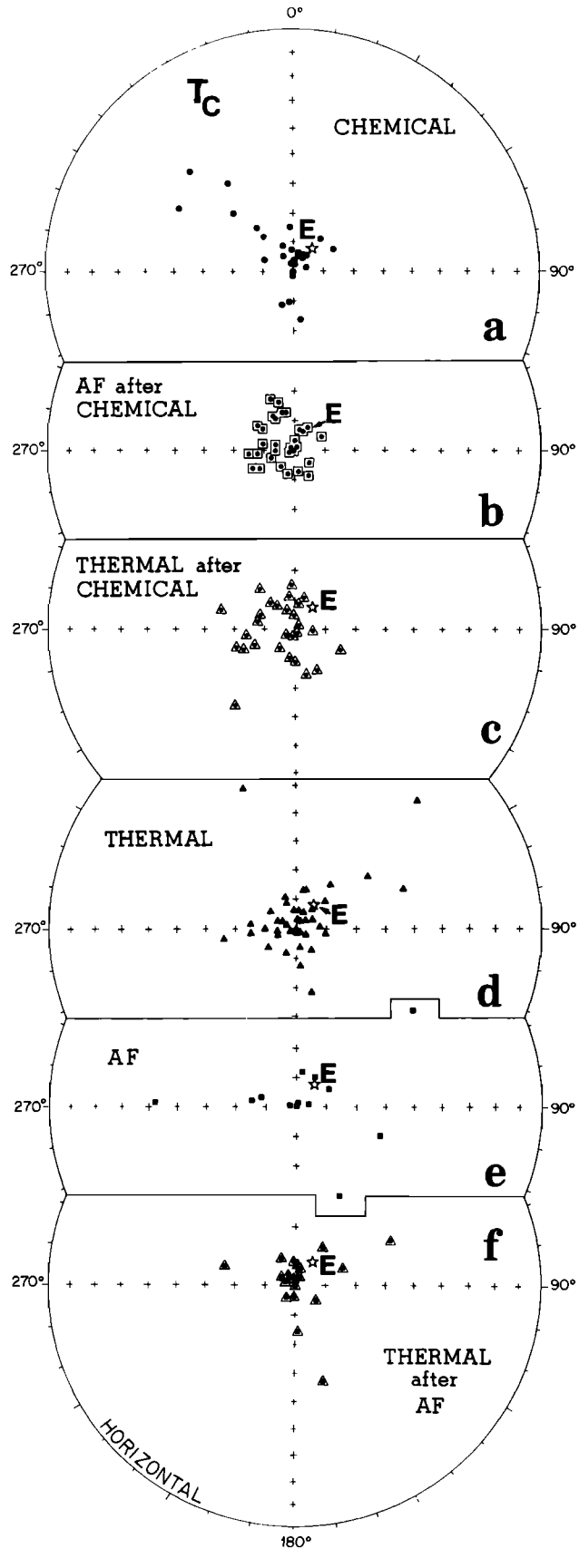


Fig. 7. Plot of all T_C specimen directions, representing magnetizations removed using a variety of techniques. Other explanations are from Figure 3.

pose thermal overprinting, forming a partial viscous thermoremanent magnetization (PVTRM). There is no evidence for postfolding thermal or volcanic events in the Mackenzie Mountains, nor for deep burial and reheating during the Tertiary. The absence of greenschist metamorphism contrains regional temperatures to an upper limit of 300°–375° [*McMechan and Price*, 1982]. Such temperatures, if present, might have been sufficient to remagnetize much of the magnetite but not the hematite [*Park and Aitken*, 1986].

In summary, the three magnetic components are very similar to those recovered from the overlying Katherine Group [*Park and Aitken*, 1986]. They have been interpreted in the same way: T_A is a primary hematite remanent magnetization, occurring mainly in detrital grains, while T_B and T_C are secondary magnetizations of prefolding and postfolding age, respectively. T_B and T_C are CRMs in two phases of hematite pigment. T_C also includes a possible PVTRM in magnetite that may reside in silicate grains.

Apparent Polar Wander Path

Poles for all components derived from the Tsezotene Formation are plotted in Figure 11, along with poles from the late Proterozoic of North America. Primary poles from sedimentary units of the Mackenzie Mountains supergroup (from the Little Dal Group (LB_A [*Park*, 1981*a*]) and Katherine Group (K_A [*Park and Aitken*, 1986]) have been interpreted as forming part of a unique APW segment for North America [*Park and Aitken*, 1986], which is separate from and younger than the 0.88–0.98 Ga Grenville Loop. The older end of this path, until now anchored by K_A, tends toward the younger end of the Grenville Track. Pole T_A from this study lies slightly north and west of K_A and is indicative of a continuing trend toward the older Grenville path.

Pole T_B is associated with poles from the Mackenzie Mountains supergroup rocks (K_B, LM_B, LL_B) that represent a pervasive red hematite pigmentation [*Morris and Aitken*, 1982; *Park*, 1984*a*, *b*; *Park and Aitken*, 1986] and with probable primary poles (CB_A, CR_A) from the "Copper cycle" [*Park*, 1984*b*, and unpublished data, 1984]. These poles are designated P2 to distinguish them from P1 poles (LB_B, LM_A, LR_A), which represent another hematite pigmentation direction confined to the Little Dal Group. Poles LM_A [*Park*, 1984*a*] and LR_A [*Park*, 1984*b*, and unpublished data, 1984] obtained from very limited data from the Mudcracked and Rusty Shale formations of the Little Dal Group, represent probable primary (or early diagenetic) magnetizations; pole LB_B from the Basinal assemblage, a clearly secondary magnetization. P2 poles have been interpreted as representing rifting that led initially to "Copper cycle" deposition and that continued through the lower Rapitan (Sayunei Formation) and perhaps into the middle Rapitan (Shezal Formation) [*Park*, 1984*b*, and unpublished data, 1984]. The P2 episode is not apparently recorded in Rapitan rocks [see *Morris*, 1977], even though sedimentation was apparently continuous in places across the "Copper cycle"–Rapitan boundary. This fact may be significant for assigning the pigmentation to "Copper cycle" times according to the following hypothesis:

"Copper cycle" sedimentation records block faulting [*Eisbacher*, 1977; *Jefferson*, 1978]. The hematite pigmentation is interpreted as recording deep flow of oxygenated groundwater through the horst blocks composed of Mackenzie Mountains supergroup strata. Meanwhile, red beds of the "Copper cycle" accumulating in grabens were acquiring a detrital/chemical remanent magnetization (CB_A, CR_A) of similar direction. Sedi-

TABLE 5. Summary of T_C Directions and Poles

Description	N	Directions D°	I°	R	k	α_{95}°	Poles °N	°W	R	K	A_{95}°	δp°	δm°
1, af, in situ	sp 15	068	+87	13.83	12	12	...	...	...	...	...	...	...
2, ch, in situ	sp 54	318	+86	52.16	29	4	...	...	...	...	...	...	...
3, th, in situ	sp 41	002	+88	39.74	32	4	...	...	...	...	...	...	...
4, af + th, in situ	sp 19	043	+88	18.37	29	6	...	...	...	...	...	...	...
5, ch + af, in situ	sp 31	309	+84	30.23	39	4	...	...	...	...	...	...	...
6, ch + th, in situ	sp 31	264	+83	29.73	24	5	...	...	...	...	...	...	...
7, total, in situ	sp 164	325	+87	157.83	26	2	...	...	...	...	...	4	4
8, Keele River 1, in situ	si 6	011	+88	5.92	60	9	67	127	...	...	...	17	17
9, Keele River 2, in situ	si 6	350	+87	5.98	238	4	69	131	...	...	...	9	9
10, Keele River 3, in situ	si 5	326	+83	4.92	50	11	74	155	...	...	...	21	22
11, Keele River 4, in situ	si 2	330	+83	1.98	...	...	74	152	...	...	...	...	...
12, Imperial River 1, in situ	si 5	256	+83	4.98	162	6	58	156	...	...	...	12	12
13, Imperial River 2, in situ	si 5	346	+87	4.97	155	6	72	133	...	...	...	12	12
14, total, in situ	si 29	317	+87	28.69	89	3	63	141	27.88	25	6	6	6
15, total, bdg	si 29	244	+88	28.15	33	5	63	136	26.43	11	8	9	10

The first six rows show the number of specimens done under different treatments. Specimen (sp) and site (si) data. The poles given for rows 8–13 are calculated from the average of the site directions; those given for rows 14 and 15 are the averages of the site poles. Row 14 is considered to be the best summary of the T_C data. Pole statistics: R is the vector resultant, K the precision, and A_{95}° the error of 95% confidence about the average of the site poles. δp°, δm° are the semi-axes of the error ellipse at probability 0.05 about the pole calculated from the average of the site directions; k is the precision of the directions. Other symbols are defined in Tables 1 and 2.

mentary rocks of the Sayunei Formation (Lower Rapitan) were deposited later in deep water and do not reveal the P2 magnetization.

It may be significant that five of the six P2 poles form a sequence. According to the proposed trend of the APW path, the youngest unit, the Redstone River Formation, has the oldest P2 magnetization, and the oldest unit, the Katherine Group, has the youngest P2 magnetization. This sequence may result from the order in which successively older rocks were uplifted sufficiently to be affected by groundwater flow. Pole T_B does not fit the sequence, for reasons not understood

Without more constraints, it is difficult to fit the P2 poles into a polar scheme. However, for the present, we connect them directly to poles LB_B and S_A at the end of the APW path

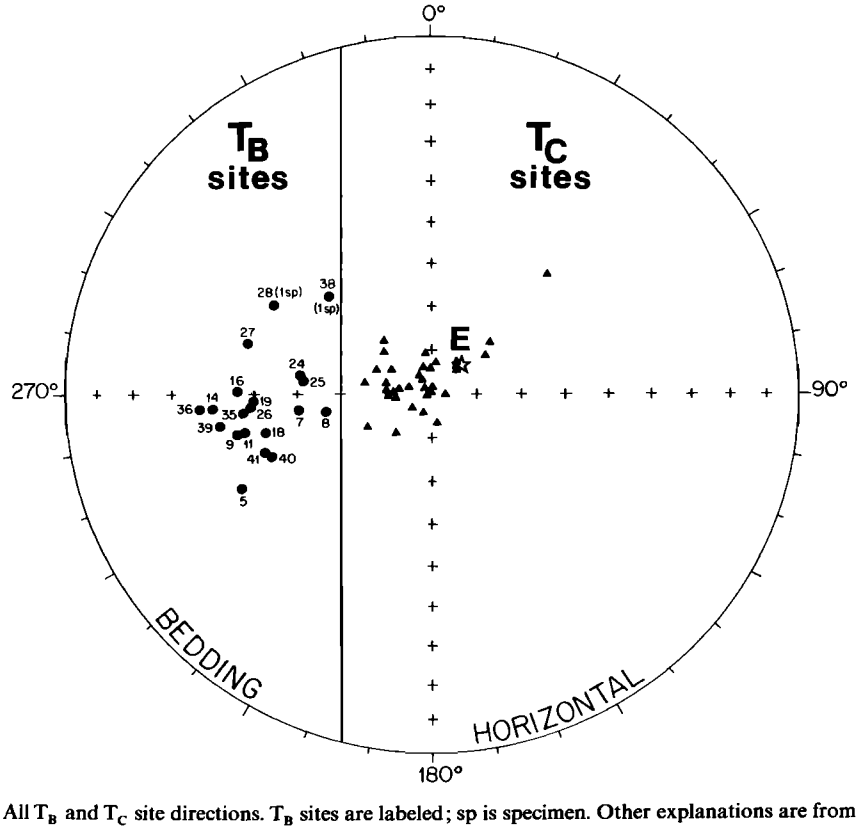


Fig. 8. All T_B and T_C site directions. T_B sites are labeled; sp is specimen. Other explanations are from Figure 3.

TABLE 6. Summary of T_B Directions and Poles

Description	N	Directions D°	I°	R	k	α_{95}°	Poles °N	°W	R	K	A_{95}°	δp°	δm°
1, ch, bdg	sp 64	262	+50	61.96	31	3	...	...	...	...	...	3	4
2, th, bdg	sp 34	263	+51	32.44	21	6	...	...	...	...	...	5	7
3, af + th, bdg	sp 14	265	+50	13.48	25	8	...	...	...	...	...	7	11
4, total, bdg	sp 112	262	+50	107.86	27	3	...	...	...	...	...	2	4
5, Keele River 1, bdg	si 6	258	+45	5.96	131	6	18	195	...	...	...	5	7
6, Keele River 2, bdg	si 3	265	+49	2.99	192	9	25	200	...	...	...	8	12
7, Keele River 3, bdg	si 4	276	+54	3.96	83	10	34	206	...	...	...	10	14
8, Keele River 4, bdg	si 2	265	+42	1.99	...	...	19	202	...	...	...	17	28
9, Imperial River 2, bdg	si 3	254	+47	2.98	102	12	19	192	...	...	...	10	16
10, total, in situ	si 18	279	+49	17.70	57	5	27	203	17.60	43	5	4	6
11, total, bdg	si 18	263	+48	17.77	73	4	23	198	17.71	58	5	3	5

The first three rows give the number of specimens (sp) done under different treatments. Site (si). For rows 10 and 11 the pole is the average of the site poles, whereas for rows 5–9 the pole is calculated from the average of the site directions. Other symbols are explained in Tables 1, 2, and 5. Row 11 is considered to be the best estimate of T_B.

TABLE 7. Summary of T_A Directions and Poles

Description	N	Directions D°	I°	R	k	α_{95}°	Poles °N	°W	δp°	δm°
1, bedding, N	12	277	+24	10.98	11	14	...	...	...	...
2, bedding, R	11	085	−24	10.65	28	9	...	...	...	...
3, in situ, all	23	272	+23	21.25	13	9	11	214	5	9
4, bedding, all	23	271	+24	21.52	15	8	12	214	5	9
5, bedding, VS, N	21	264	+23	20.26	27	6	...	...	...	...
6, bedding, VS, R	11	075	−27	10.24	13	13	...	...	...	...
7, bedding, VS, all	21	263	+24	20.37	32	6	...	...	...	...

Polarity of direction: normal (N), reverse (R). Other symbols are explained in Tables 2 and 5. Rows 5–7 were obtained from vector substraction (VS). They are not considered to be reliable enough to include with rows 1–4 and are therefore given for comparison. Some VS specimens have N and R directions; these were averaged together before the row 7 average was determined. Row 4 is considered to be the best estimate of T_A.

TABLE 8. Application of Tilt Test

Description	N	Precision k of Averages HOR (k_1)	BDG (k_2)	k_2/k_1	95% Confidence Limit*	Significance of Test
T_A Imperial River	6 sp	5.88	6.75	1.14	2.97	NO
T_A all sites	23 sp	12.56	14.87	1.18	1.66	NO
T_B	18 si	53.36	59.86	1.12	1.78	NO
T_C Imperial River†	10 si	118.30	20.87	(5.66)	3.13‡	(YES)
T_C all sites	29 si	88.84	32.82	(2.71)	1.57	(YES)

k is given before (HOR) and after (BDG) correction for bedding. N is the number of specimens (sp) or sites (si). Parentheses refer to the ratio of k_1/k_2 (negative fold test).

*Values are from *McElhinny* [1964].

†Classical "fold test" [*Graham*, 1949].

‡99% confidence limit.

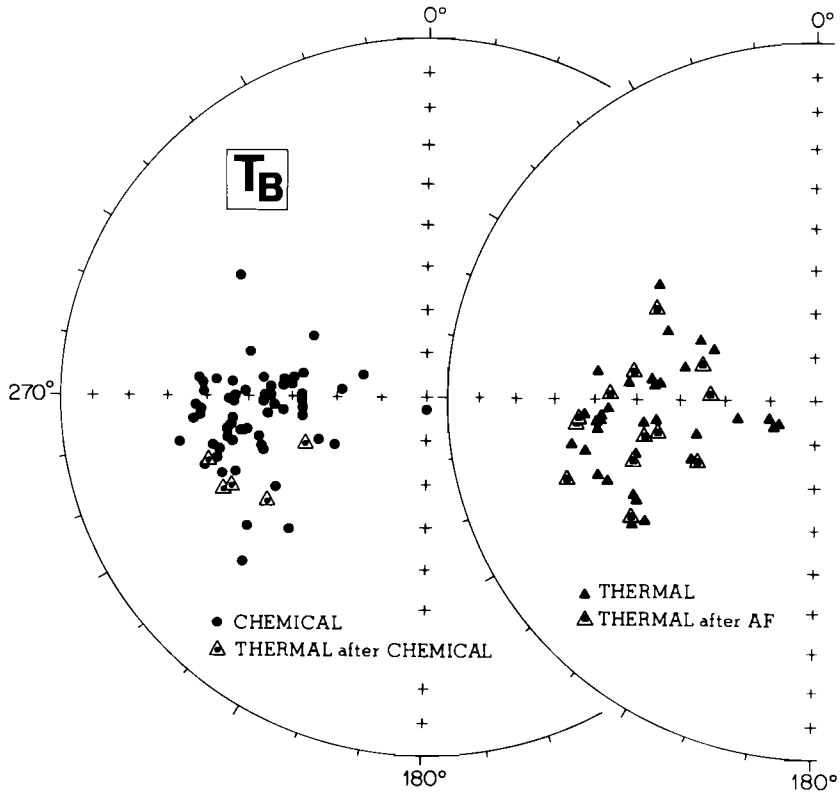


Fig. 9. Specimen directions from T_B magnetizations removed by a variety of techniques. Other explanations are from Figure 3.

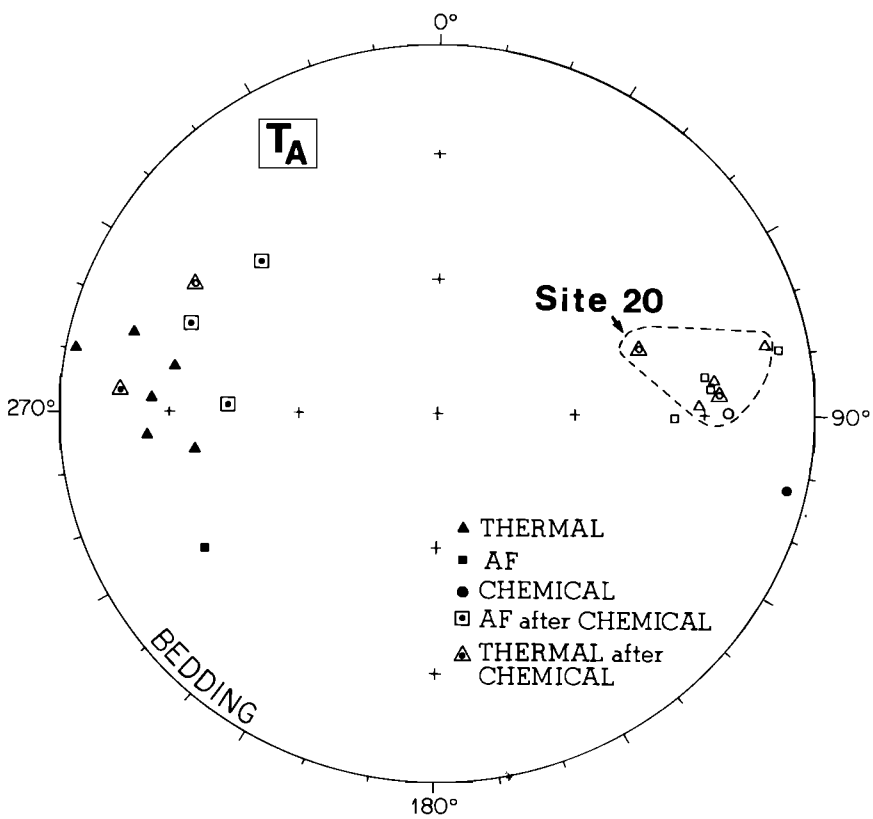


Fig. 10. All T_A specimen directions, representing magnetizations removed by a variety of techniques. Other explanations are from Figure 3.

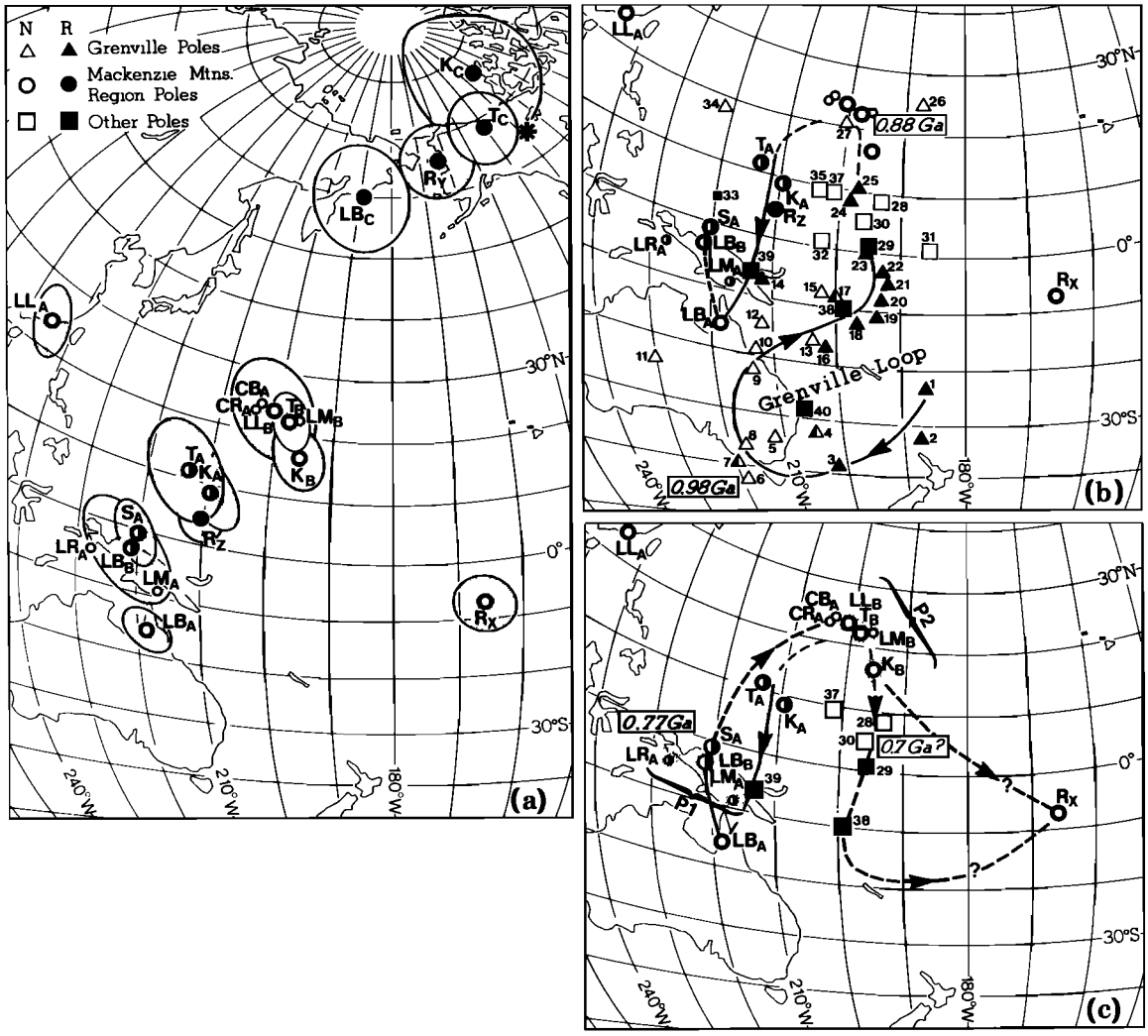


Fig. 11. After *Park and Aitken* [1986]. Paleomagnetic poles from predominantly late Precambrian rocks of North America (1000–700 Ma). (*a*) Poles derived from the Mackenzie Mountains region together with their errors ($\delta p°$, $\delta m°$; or $A_{95}°$). (*b*) Most relevant poles from North America showing the Grenville Loop and a proposed new loop [*Park and Aitken*, 1986] drawn through poles of the Mackenzie Mountains region. (*c*) Proposed new polar path drawn through late Precambrian Mackenzie Mountains poles and other possibly relevant poles. The asterisk denotes the sampling region. Open symbols denote normal poles, and solid symbols denote reverse. Small symbols denote limited data. Mackenzie Mountains poles are labeled with letters. A subscript "A" denotes a probable primary pole; subscripts "B" and "C" secondary poles. Shaded poles are derived from magnetizations carried by red hematite pigment. The poles shown are as follows. Mackenzie Mountains poles: CB_B, Basal Sandy member ("Copper cycle" [*Park*, 1984*b*; and unpublished data, 1984]); CR_B, Redstone River Formation ("Copper cycle" [*Park*, 1984*b*; and unpublished data, 1984]); K_A, K_B, K_C, Katherine Group [*Park and Aitken*, 1986]; LB_A, LB_B, LB_C, Basinal Sequence (Little Dal Group [*Park*, 1981*a*]); LL_A, LL, LL_C, lavas (Little Dal Group [*Morris and Park*, 1981; *Morris and Aitken*, 1982]); LM, LM_B, Mudcracked formation (Little Dal Group [*Park*, 1984*a*]); LR, Rusty Shale formation (Little Dal Group [*Park*, 1984*b*; and unpublished data, 1984]); R_X, R_Y, R_Z, Rapitan Group [*Morris*, 1977]; S_A sills in Tsezotene Formation [*Morris and Park*, 1981; *Park* 1981*b*]; T_A, T_B, T_C, Tsezotene Formation (this paper). Grenville Structural Province poles: 1, Mealy Mountains dykes, A component [*Park and Emslie*, 1983]; 2, 3, Thanet gabbro A_2 and A_1 components [*Buchan*, 1978]; 4, Grenville diorites, Quebec (322 [*Irving et al.*, 1972]); 5, Mealy Mountains dykes, B component [*Park and Emslie*, 1983]; 6, Morin anorthosite complex (482 [*Irving et al.*, 1974]); 7, Allard Lake anorthosite [*Hargraves and Burt*, 1967]; 8, Haliburton intrusions, Hb_1 [*Buchan and Dunlop*, 1976]; 9, Whitestone diorite, WZ (503 [*Ueno et al.*, 1975]); 10, Lac St.-Jean anorthosite, normal [*Buchan et al.*, 1983]; 11, Magnetawan metasediments [*McWilliams and Dunlop*, 1975]; 12, Wilberforce pyroxenite (457 [*Palmer and Carmichael*, 1973]); 13, Whitestone anorthosite, WW (500 [*Ueno et al.*, 1975]); 14, St.-Urbain anorthosite (409 [*Robertson and Roy*, 1979]); 15, Indian Head anorthosite [*Murthy and Rao*, 1976]; 16, Seal Lake Group, B [*Roy and Fahrig*, 1973; *Park and Roy*, 1979]; 17, Umfraville gabbro [*Palmer et al.*, 1979]; 18, Frontenac Axis dykes (363 [*Park and Irving*, 1972]); 19, Umfraville gabbro [*Symons*, 1978]; 20, Lac St.-Jean anorthosite, reverse [*Buchan et al.*, 1983]; 21, Whitestone anorthosite, WY (502 [*Ueno et al.*, 1975]); 22, Haliburton intrusions, Hb_2 [*Buchan and Dunlop*, 1976]; 23, Morin anorthosite complex (483 [*Irving et al.*, 1974]); 24, Grenville Front anorthosites (459 [*Palmer and Carmichael*, 1973]); 25, Michael gabbro (352 [*Fahrig and Larochelle*, 1972]); 26, Haliburton intrusions, Hb_3 [*Buchan and Dunlop*, 1976]; 27, Thanet gabbro, B [*Buchan*, 1978]; 34, Tudor gabbro (458 [*Palmer and Carmichael*, 1973]). Other North American poles: 28, Franklin diabases, normal (27 sites [*Fahrig et al.*, 1971; *Fahrig and Schwarz*, 1973; *Park*, 1974; *Christie and Fahrig*, 1983]); 29, Franklin diabases, reverse (51 sites [*Fahrig et al.*, 1971; *Robertson and Baragar*, 1972; *Fahrig and Schwarz*, 1973; *Park*, 1974; *Palmer and Hayatsu*, 1975; *Park*, 1981*c*; *Christie and Fahrig*, 1983]); 30, Spokane Formation, overprint [*Vitorello and Van Der Voo*, 1977]; 31, Rama diabase, Arizona (391, [*Larson et al.*, 1973]); 32, Uinta Mountain Group, lower sites [*Bressler*, 1981]; 33, Uinta Mountain Group, upper site [*Bressler*, 1981]; 35, 36, Mount Nelson Formation, secondary and primary? [*Morris and McMechan*, 1983]; 37, 38, Natkusiak basalts, normal and reverse [*Palmer and Hayatsu*, 1975; *Palmer et al.*, 1983]; 39, Reynolds Point Formation [*Palmer et al.*, 1983]; 40, Borden dykes [*Christie and Fahrig*, 1983]. Lead numbers in parentheses above refer to the Ottawa pole catalog list [*Irving and Hastie*, 1975].

for the Mackenzie Mountains supergroup [*Park and Aitken*, 1986] (Figure 11). In the forward sense there are two obvious possibilities: the P2 poles may connect with the Franklin poles or with the possible primary R_X pole from the Sayunei Formation [*Park and Aitken*, 1986].

The T_C pole from the Tsezotene Formation is post-Paleocene in age and is probably related to other poles from the region: K_C [*Park and Aitken*, 1986], R_Y [*Morris*, 1977], and LB_C [*Park*, 1981*a*]. The latter poles are of both prefolding and postfolding age and are derived from magnetizations residing in red pigment formed of hematite or related minerals [*Park and Aitken*, 1986]. We therefore suggest that T_C, though postfolding, relates to the Laramide Orogeny.

Poles from the Mackenzie Mountains do not appear to be affected by structural rotations [*Park and Aitken*, 1986], commonly found associated with thrust sheets and detached folds within fold and thrust belts. Additional support for this conclusion is found by comparing T_B poles (three sites; 19°N, 192°W) from the Imperial River Canyon region with T_B poles (15 sites; 24°N, 199°W) from the Keele River region. They agree within error, though the regions are located on separate structural blocks. Also, T_B is closely associated with other P2 poles, which together with T_B come from four thrust sheets. In addition, relative rotation between the fold belt and the craton are similarly excluded [*Park and Aitken*, 1986]. Evidence for this latter assertion hinges primarily on the close correspondence between a pole (LB_A) from the lower Little Dal Group of the Mackenzie Mountains and a pole (39) from the geologically-correlated Reynolds Point Formation from Victoria Island on the craton [*Park and Aitken*, 1986]. Weaker evidence comes from the postulated relationship between the Franklin poles (28, 29) from the craton and the P2 poles from the Mackenzie Mountains [*Park and Aitken*, 1986].

Acknowledgments. We are grateful for the field assistance of D. G. F. Long and for his help and advice after the field sampling. We thank the following students for their help in the field: Kathy Black, Kevin Sharman, Robin Sandau, and Sylvain Savard. Sylvain Savard helped to prepare samples in the laboratory. Jean Hastie, Sylvain Savard, and Elizabeth Tanczyk performed some of the measurements. One of us (J. K. P.) thanks P. Lapointe and J. L. Roy of the Earth Physics Branch for useful discussions of the work at critical points. Finally, we are grateful to J. L. Roy for reviewing the manuscript. Contribution 1208 from the Earth Physics Branch.

REFERENCES

Aitken, J. D., Stratigraphy and sedimentology of the upper Proterozoic Little Dal Group, Mackenzie Mountains, Northwest Territories, Proterozoic Basins of Canada, edited by F. H. A. Campbell, *Geol. Surv. Pap. Geol. Surv. Can., 81-10*, 47–71, 1981.

Aitken, J. D., and D. G. Cook, Carcajou Canyon map area, District of Mackenzie, Northwest Territories, *Geol. Surv. Pap. Geol. Surv. Can., 74-13*, 1–28, 1974.

Aitken, J. D., D. G. F. Long, and M. A. Semikhatov, Progress in Helikian stratigraphy, Mackenzie Mountains, *Geol. Surv. Pap. Geol. Surv. Can., 78-1A*, 481–484, 1978.

Aitken, J. D., D. G. Cook, and C. J. Yorath, Upper Ramparts River and Sans Sault Rapids Map-Areas (106G, H), District of Mackenzie (with maps 1452A and 1453A) *Mem. Geol. Surv. Can., 388*, 1–48, 1982.

Armstrong, R. L., G. H. Eisbacher, and P. D. Evans, Age and stratigrahic-tectonic significance of Proterozoic diabase sheets, Mackenzie Mountains, northwestern Canada, *Can. J. Earth Sci., 19*, 316–323, 1982.

Bressler, S. L., Preliminary paleomagnetic poles and correlation of the Proterozoic Uinta Mountain Group, Utah and Colorado, *Earth Planet. Sci. Lett., 55*, 53–64, 1981.

Buchan, K. L., Magnetic overprinting in the Thanet gabbro complex, Ontario, *Can. J. Earth Sci., 15*, 1407–1421, 1978.

Buchan, K. L., and D. J. Dunlop, Paleomagnetism of the Haliburton intrusions: Superimposed magnetizations, metamorphism and tectonics in the late Precambrian, *J. Geophys. Res., 81*, 2951–2967, 1976.

Buchan, K. L., W. F. Fahrig, G. N. Freda, and R. A. Frith, Paleomagnetism of the Lac St-Jean anorthosite and related rocks, Grenville Province, Quebec, *Can. J. Earth Sci., 20*, 246–258, 1983.

Christie, K. W., and W. F. Fahrig, Paleomagnetism of the Borden dykes of Baffin Island and its bearing on the Grenville Loop, *Can. J. Earth Sci., 20*, 275–289, 1983.

Dallmeyer, R. D., and J. F. Sutter, Acquisitional chronology of remanent magnetization along the "Grenville polar path": Evidence from $^{40}Ar/^{39}Ar$ ages of hornblende and biotite from the Whitestone diorite, Ontario, *J. Geophys. Res., 85*, 3177–3186, 1980.

Donaldson, J. A., Possible correlations between Proterozoic strata of the Canadian Shield and North American Cordillera, in *Belt Symposium*, vol. 1, pp. 61-75, University of Idaho, Moscow, 1973.

Douglas, R. J. W., Geological map of Canada, map 1250 A, Accompanies *Econ. Geol. Rep. Geol. Surv. Can., 1*, 5th ed., 1970.

Eisbacher, G. H., Tectonostratigraphic framework of the Redstone Copper Belt, District of Mackenzie, Report of Activities, part A, *Geol. Surv. Pap. Geol. Surv. Can., 77-1A*, 229–234, 1977.

Eisbacher, G. H., Re-definition and subdivision of the Rapitan Group, Mackenzie Mountains, *Geol. Surv. Pap. Geol. Surv. Can., 77-35*, 1–21, 1978.

Eisbacher, G. H., Sedimentary tectonics and glacial record in the Windermere Supergroup, Mackenzie Mountains, northwestern Canada, *Geol. Surv. Pap. Geol. Surv. Can., 80-27*, 1–40, 1981.

Evenchick, C. A., R. R. Parrish, and H. Gabrielse, Precambrian gneiss and late Proterozoic sedimentation in north-central British Columbia, *Geology, 12*, 233–237, 1984.

Fahrig, W. F., and A. Larochelle, Paleomagnetism of the Michael gabbro and possible evidence of the rotation of Makkovik Subprovince, *Can. J. Earth Sci., 9*, 1287–1296, 1972.

Fahrig, W. F., and E. J. Schwarz, Additional paleomagnetic data on the Baffin diabase dikes and a revised Franklin pole, *Can. J. Earth Sci., 10*, 576–581, 1973.

Fahrig, W. F., E. Irving, and G. D. Jackson, Paleomagnetism of the Franklin diabases, *Can. J. Earth Sci., 8*, 455–467, 1971.

Gabrielse, H., Younger Precambrian of the Canadian Cordillera, *Am. J. Sci., 272*, 521–536, 1972.

Gabrielse, H., S. L. Blusson, and J. A. Roddick, Geology of Flat River, Glacier Lake, and Wrigley Lake map-areas, District of Mackenzie and Yukon Territory, 1, General geology, structural geology, and economic geology, *Mem. Geol. Surv. Can., 366*, 1–153, 1973.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, *J. Geophys. Res., 54*, 131–167, 1949.

Hargraves, R. B., and D. W. Burt, Paleomagnetism of the Allard Lake anorthosite suite, *Can. J. Earth Sci., 4*, 357–369, 1967.

Henshaw, Jr., P. C., and R. T. Merrill, Magnetic and chemical changes in marine sediments, *Rev. Geophys., 18*, 483–504, 1980.

Irving, E., and J. Hastie, *Catalogue of Paleomagnetic Directions and Poles, Geomagn. Ser.*, vol. 3, Earth Physics Branch, Ottawa, Ont., 1975.

Irving, E., J. K. Park, and J. L. Roy, Paleomagnetism and the origin of the Grenville Front, *Nature, 236*, 344–346, 1972.

Irving, E., J. K. Park and R. F. Emslie, Paleomagnetism of the Morin complex, *J. Geophys. Res., 79*, 5482–5490, 1974.

Jefferson, C. W., Stratigraphy and sedimentology, upper Proterozoic Redstone Copper Belt, Mackenzie Mountains, Northwest Territories—A preliminary report, in *Mineral Industry Report for 1975, Econ. Geol. Ser. 1978–5*, pp. 157–169, Northwest Territories, Indian and Northern Affairs, Ottawa, 1978.

Jefferson, C. W., and J. C. Ruelle, The late Proterozoic Redstone Copper Belt, Mackenzie Mountains, NWT, in *Mineral Deposits of the Northern Cordillera*, edited by J. A. Morin, Canadian Institute of Mining and Metallurgy, Montreal, Quebec, in press, 1986.

King, P. B., Tectonic map of North America, U.S. Geol. Surv., Reston, Va., 1969.

Larson, E. E., R. Reynolds, and R. Hobbit, New virtual and paleomagnetic pole positions from isotopically dated Precambrian rocks in Wyoming, Montana and Arizona: Their significance in establishing a North American apparent polar wandering path, *Geol. Soc. Am. Bull., 84*, 3231–3248, 1973.

McElhinney, M. W., Statistical significance of the fold test in palaeomagnetism, *Geophys. J. R. Astron. Soc., 8*, 338–340, 1964.

McMechan, M. E., and R. A. Price, Superimposed low-grade metamorphism in the Mount Fisher area, southeastern British Columbia—Implications for the East Kootenay orogeny, *Can. J. Earth Sci., 19*, 476–489, 1982.

McWilliams, M. O., and D. J. Dunlop, Precambrian paleomagnetism: Magnetizations reset by the Grenville orogeny, *Science, 190*, 269–271, 1975.

Morris, W. A., Paleolatitude of glaciogenic upper Precambrian Rapitan Group and the use of tillites as chronostratigraphic marker horizons, *Geology, 5*, 85–88, 1977.

Morris, W. A., and J. D. Aitken, Paleomagnetism of the Little Dal lavas, Mackenzie Mountains, Northwest Territories, Canada, *Can. J. Earth Sci., 19*, 2020–2027, 1982.

Morris, W. A., and M. E. McMechan, Paleomagnetism of the middle Proterozoic Mount Nelson Formation: Evidence for a regional remagnetization event in the late Precambrian of the Cordillera, *Can. J. Earth Sci., 20*, 561–567, 1983.

Morris, W. A., and J. K. Park, Correlation of upper Proterozoic strata in the Cordillera: Paleomagnetism of the Tsezotene sills and the Little Dal lavas, Proterozoic Basins of Canada, edited by F. H. A. Campbell, *Geol. Surv. Pap. Geol. Surv. Can., 81-10*, 73–78, 1981.

Murthy, G. S., and K. V. Rao, Paleomagnetism of the Steel Mountains and Indian Head anorthosites from western Newfoundland, *Can. J. Earth Sci., 13*, 75–83, 1976.

Palmer, H. C., and C. M. Carmichael, Paleomagnetism of some Grenville Province rocks, *Can. J. Earth Sci., 10*, 1175–1190, 1973.

Palmer, H. C., and A. Hayatsu, Paleomagnetism and K-Ar dating of some Franklin lavas and diabases, Victoria Island, *Can. J. Earth Sci., 12*, 1439–1447, 1975.

Palmer, H. C., A. Hayatsu, C. E. Waboso, and S. Pullen, A paleomagnetic and K-Ar study of the Umfraville gabbro, Ontario, *Can. J. Earth Sci., 16*, 459–471, 1979.

Palmer, H. C., W. R. A. Baragar, M. Fortier, and J. H. Foster, Paleomagnetism of late Proterozoic rocks, Victoria Island, Northwest Territories, Canada, *Can. J. Earth Sci., 20*, 1456–1469, 1983.

Park, J. K., Acid leaching of red beds and the relative stability of the red and black magnetic components *Can. J. Earth Sci., 7*, 1086–1092, 1970.

Park, J. K., Paleomagnetism of miscellaneous Franklin and Mackenzie diabases of the Canadian Shield and their adjacent country rocks, *Can. J. Earth Sci., 11*, 1012–1017, 1974.

Park, J. K., Analysis of the multicomponent magnetization of the Little Dal Group, Mackenzie Mountains, Northwest Territories, Canada, *J. Geophys. Res., 86*, 5134–5146, 1981*a*.

Park, J. K., Paleomagnetism of the late Proterozoic sills in the Tsezotene Formation, Mackenzie Mountains, Northwest Territories, Canada, *Can. J. Earth Sci., 18*, 1572–1580, 1981*b*.

Park, J. K., Paleomagnetism of basic instrusions from the Brock Inlier, Northwest Territories, Canada, *Can. J. Earth Sci., 18*, 1637–1641, 1981*c*.

Park, J. K., Paleomagnetism for geologists, *Geosci. Can., 10*, 180–188, 1983.

Park, J. K., Paleomagnetism of the Mudcracked formtion of the Precambrian Little Dal Group, Mackenzie Mountains, Northwest Territories, Canada, *Can. J. Earth Sci., 21*, 371–375, 1984*a*.

Park, J. K., Geological implications of chemical overprinting in the Precambrian Mackenzie Mountains supergroup (MMs), Northwest Territories, paper presented at the Geological Association of Canada and Mineralogical Association of Canada Joint Annual Meeting, London, Ont., 1984*b*.

Park, J. K., and J. D. Aitken, Paleomagnetism of the Katherine Group in the Mackenzie Mountains: Implications for post-Grenville (Hadrynian) pole wander, *Can. J. Earth Sci.*, in press, 1986.

Park, J. K., and R. F. Emslie, Paleomagnetic history of the Mealy dykes of Labrador, Canada, Can. J. Earth Sci., *20*, 1818–1833, 1983.

Park, J. K., and E. Irving, Magnetism of dikes of the Frontenac Axis, *Can. J. Earth Sci., 9*, 763–765, 1972.

Park, J. K., and J. L. Roy, Further paleomagnetic results from the Seal Lake Group igneous rocks, Labrador, *Can. J. Earth Sci., 16*, 895–912, 1979.

Robertson, P. B., and J. L. Roy, Shock-diminished paleomagnetic remanence at the Charlevoix impact structure, Quebec, *Can. J. Earth Sci., 16*, 1842–1856, 1979.

Robertson, W. A., and W. R. A. Baragar, The petrology and paleomagnetism of the Coronation sills, Can. J. Earth Sci., *9*, 123–140, 1972.

Roy, J. L., and W. F. Fahrig, The paleomagnetism of Seal and Croteau rocks from the Grenville Front, Labrador: Polar wandering and tectonic implications, *Can. J. Earth Sci., 10*, 1279–1301, 1973.

Roy, J. L., and J. K. Park, The magnetization process of certain red beds: Vector analysis of chemical and thermal results, *Can. J. Earth Sci., 11*, 437–471, 1974.

Roy, J. L., W. A. Robertson, and C. Keeping, Magnetic "field free" spaces for paleomagnetism, rock magnetism, and other studies, *Can. J. Earth Sci., 6*, 1312–1316, 1969.

Roy, J. L., E. Sanders, and J. Reynolds, Un four électrique pour l'étude des propriétés magnétiques des roches, *Publ. Dir. Phys. Globe, 42*, 229–237, 1972.

Roy, J. L., J. Reynolds, and E. Sanders, An alternating field demagnetizer for rock magnetism studies, *Publ. Earth Phys. Branch Can., 44*(3), 37–45, 1973.

Stewart, J. H., Initial deposits in Cordilleran geosyncline: Evidence of a late Precambrian (<850 m.y.) continental separation, *Geol. Soc. Am. Bull., 83*, 1345–1360, 1972.

Symons, D. T. A., Paleomagnetism of the 1180 Ma Grenvillian Umfraville gabbro, Ontario, *Can. J. Earth Sci., 15*, 956–962, 1978.

Ueno, H., E. Irving, and R. H. McNutt, Paleomagnetism of the Whitestone anorthosite and diorite, the Grenville polar track, and relative motions of the Laurentian and Baltic Shields, *Can. J. Earth Sci., 12*, 209–226, 1975.

Vitorello, I., and R. Van Der Voo, Late Hadrynian and Helikian pole positions from the Spokane Formation, Montana, *Can. J. Earth Sci., 14*, 67–73, 1977.

Young, G. M., Correlation of middle and upper Proterozoic strata of the northern rim of the North Atlantic craton, *Trans. R. Soc. Edinburgh, 70*, 323–336, 1979.

Young, G. M., C. W. Jefferson, G. D. Delaney, and G. M. Yeo, Middle and late Proterozoic evolution of the northern Canadian Cordillera and Shield, *Geology, 7*, 125–128, 1979.

J. D. Aiken, Institute of Sedimentary and Petroleum Geology, Geological Survey of Canada, 3303 33rd Street, N. W., Calgary, Alberta, Canada T2L 2A7.

J. K. Park, Division of Seismology and Geomagnetism, Earth Physics Branch, 1 Observatory Crescent, Ottawa, Ontario, Canada K1A 0Y3.

(Received May 13, 1985;
revised October 10, 1985;
accepted December 3, 1985.)